AUTUMN MID SEMESTER EXAMINATION 2024

School of Computer Engineering

Kalinga Institute of Industrial Technology, Deemed to be University

High Performance Computing [CS30005]

Time: 90 minutes **5th Semester** **Full Marks: 20**

Answer **any four questions**, including **Question 1**, which is **compulsory**.

Q1 – This question is compulsory.

a. A program has an effective CPI of 2.5 on a CPU. If a compiler optimization reduces the number of instructions by 20% while increasing the effective CPI by 10%, what is the new execution time as a percentage of the original? [1]

b. For a MIPS jump instruction (e.g., J), how far can it jump in terms of the number of instructions on either side of the Program Counter (PC)? Specify the maximum number of instructions it can jump both backward and forward. [1]

c. Why does using a value loaded from memory with a lw (load word) instruction in a 5-stage pipelined MIPS CPU immediately after the `lw` necessitate at least one cycle of stall? Provide a brief explanation. [1]

d. Provide an example of a Write After Write (WAW) hazard in a MIPS pipeline. Briefly explain the potential issue this hazard can cause in terms of program execution. [1]

e. A 7-stage pipeline processor has the uniform stage delays as 80ns. Pipeline registers are used between the stages and have a delay of 2 ns each. Calculate the cycle time for pipelined processor and non-pipelined processor. [1]

Q2

a. Consider a program consisting of n classes of instructions each with a CPI (Cycles Per Instruction) of CPI_1, CPI_2 … CPI_n. The instructions of each class occur in the program with percentages P_1, P_2 … P_n. The program is executed on a k-core CPU with a clock frequency of **f**. The total number of instructions in the program is I, and S% of the program is sequential. Derive an expression for the performance of the program in terms of the given parameters. [2]

b. Consider a 2-bit branch predictor that is initially in the 01 state. The table below shows a sequence of branch decisions. Fill in the following table with the predictor's behavior for each branch instruction. The columns are:
State: The current state of the 2-bit predictor before the branch instruction.
Predicted Decision: The decision predicted by the branch predictor based on its current state.
Actual Decision: The actual outcome of the branch (Taken or Not Taken).
New State: The updated state of the predictor after the actual decision is known.
Correct/Not Correct: Indicate whether the predicted decision was correct or not.

State	Predicted Decision	Actual Decision	New State	Correct/Not Correct
01	?	Not Taken	?	?
?	?	Taken	?	?
?	?	Not Taken	?	?
?	?	Taken	?	?
?	?	Taken	?	?
?	?	Not Taken	?	?

[3]

Q3

a. Below is a MIPS program with 10 instructions, including a branch. Assume that without branch prediction, there is a stall of two cycles when the branch instruction is executed.

```
add   R9,   R9,   R10
sub   R11,  R12,  R9
lw    R14,  0(R11)
```

```
        and     R24,  R25,  R8
        or      R23,  R11,  R21
        beq     R8,   R7,   Label
        addi    R21,  R21,  1
        sw      R14,  4(R23)
Label:  add     R16,  R17,  R18
```

How many delay slots need to be filled to avoid any stalls when the branch instruction (`beq`) is executed? Which instructions from the program can be moved to fill these delay slots, ensuring no change in the program's functionality and avoiding any stalls? You are not allowed to use `NOP` as a delay slot instruction. Assume that operand forwarding is enabled. [2]

b. Given the following 32-bit hexadecimal MIPS instructions, decode all parts of the instruction, including opcode, register fields, function code (if applicable), and immediate or target address (if applicable). Write out the complete assembly instruction for each.

0x01398820 (instruction: `add`)
0x8C130004 (instruction: `lw`)
0x0800000A (instruction: `j`) [3]

Q4

a. Consider the followingcode and identify 2 RAW,2WAR, and 2WAW hazards. It is being run on a 5-stage pipeline where the last stage is used for register write and there is no operator forwarding.

```
add     R8,   R9,   R10
sub     R11,  R12,  R13
lw      R9,   0(R8)
and     R14,  R9,   R15
or      R24,  R25,  R14
addi    R12,  R12,  1
sub     R12,  R12,  R8
xor     R12,  R11,  R10
sw      R11,  4(R10)
nor     R10,  R9,   R13
or      R9,   R1,   R9
```

[2]

b. Derive an expression for pipeline speed up, throughput, and efficiency with k-stages. [3]

Q5.

a. List four key differences between CISC (Complex Instruction Set Computer) and RISC (Reduced Instruction Set Computer) architectures. [2]

b. Consider the following MIPS program.

```
lw      R9,   0(R10)
add     R11,  R9,   R12
sub     R13,  R11,  R14
and     R15,  R13,  R16
or      R17,  R15,  R8
sw      R17,  4(R10)
add     R9,   R9,   R11
```

Without operand forwarding, find the number of cycles needed for execution. Clearly show yourwork using a timeline.

With operand forwarding, find the number of cycles needed for execution. Show a timeline and use arrows to show forwarding paths.

[3]